Haowen Xu, Ph.D.

CONTACT INFORMATION

Research Fellow (Level B)
School of Built Environment
UNSW Sydney
Kensington, NSW, Australia
Work Email: haowen.xu.phd@gmail.com
Official Email: haowen.xu1@unsw.edu.au

Research Scientist
Computational Sciences and Engineering Division
Oak Ridge National Laboratory (ORNL)
Oak Ridge, Tennessee, USA
Official Email: xuh4@ornl.gov

Adjunct Associate Professor
Department of Industrial and Systems Engineering
University of Tennessee, Knoxville
Knoxville, Tennessee, USA

WEB PRESENCE Personal Website – Google Scholar – LinkedIn – ResearchGate – ORCID

PROFILE

Synopsis of Credentials:

- Six years of post-Ph.D. research experience with:
  - 1 Significant Event Award (SEA) from ORNL in 2020
  - 3 Invention Disclosures (Nonprovisional U.S. Patent) from 2019-2025
  - 2 US Patents in Intelligent Transportation Applications (Application Number - US20250131819A1 & US20250128708A1)
  - Holds **Permanent Residency in Australia**; eligible to work without visa sponsorship
- 25 peer-reviewed journal publications and over 20 peer-reviewed conference publications, presentations, white papers, and technical reports
  - 8 First/corresponding-author publications in premier journals such as the IEEE Transactions on Intelligent Transportation System (T-ITS), IEEE Transactions on Intelligent Vehicles (T-IV), Science of the Total Environment (STOTEN), Environmental Modelling & Software (EMS), Journal of Environmental Management (JEM), and Journal of Hydrology (JoH).
- Received best research/poster awards from world-renowned professional associations, such as the International Building Performance Simulation Association (IBPSA), Institute of Electrical and Electronics Engineers (IEEE), and American Water Resources Association (AWRA)
- Co-Principal Investigator (CoPI) of national projects funded by the U.S. Department of Energy's Advanced Research Projects Agency-Energy (ARPA-E) and Nuclear Safety Research and Development (NSRD) program
- Research outcomes have been featured and highlighted by media such as CNN, and professional association, such as the American Water Resources Association (AWRA) Impact and International Association for Hydro-Environment Engineering and Research (IAHR) Hydro-link magazine
- *Domain expertise* in urban informatics, smart and sustainable cities, water resources management, natural hazard mitigation, and erosion and sediment transport
- Serving as an *Adjunct Faculty* at the University of Tennessee Knoxville (UTK), and has advised and co-advised two Ph.D. students

RESEARCH EXPERTISE

Focus Areas:

- AI agents for smart cities using Large Language Models (LLMs) and Prompt Engineering.
- Generative AI for Urban and Environmental Modeling.
- Intelligent Transportation System (ITS) for smart mobility management.
- Visual Analytics for explorative analysis and decision support.
- Cyberinfrastructure, Digital Twin (DT), and Cyber Physical system (CPS) for developing smart cities, buildings, grids, and mobility systems.
- Virtual Reality (VR) and Augmented Reality (AR) for advanced and immersive scientific visualization.
- Urban informatics for decarbonization and sustainable economy.
- Hydroinformatics and Environmental Informatics for water hazards mitigation.

Technical Skills:

- Expertise in software engineering, design and architecture patterns (e.g., MVC and MVVM), DevOps (Docker and Kubernetes), and a wide variety of frameworks and software stacks for developing web/mobile/cloud/edge-computing application and large-scale Cyberinfrastructure (CI).
- Proficient in utilizing pre-trained Large Language Models (LLMs), including ChatGPT APIs, Ollama, and Retrieval-Augmented Generation (RAG), alongside experience with relational, graph, and vector databases.
- Proficient in developing Generative AI models and Agentic AI architecture, including VAEs, Transformers, and Diffusion Models.
- Experienced in multiple Python libraries for data science and deep learning, including Pandas, NumPy, SciPy, Keras, and PyTorch.
- Expertise in prominent programming languages and libraries (e.g., D3JS and WebGL) for developing customized visual encodings, visualizations, and visual analytics methdology for exploring complex scientific data and supporting explainable Artificial Intelligence (AI).
- Expertise in various GIS software and spatial analytics tools, such as ArcGIS, ArcPy, QGIS, and GDAL.
- Expertise in various 3D modeling software and game engines for developing VR and AR applications, including BlenderGIS, AutoDesk Maya, ThreeJS, and Unity 3D.
- Experienced with popular hydrologic and environmental simulation models, such as HEC-HMS, HEC-RAS, HEC-SSP, Height Above Nearest Drainage (HAND), SWAT, and RUSLE.
- Experstise in national hydrological and environmental datasets, such as the USGS National Hydrography Dataset, EPA Stream-Catchment (StreamCat) Dataset, and USGS SSURGO soil datasets.
- Experienced with popular traffic simulation software that include SUMO and VISSIM.

EDUCATION

Ph.D., Civil Engineering 2015 - 2019
The University of Iowa, Iowa City, IA
Cumulative GPA: 3.90/4.00
Dissertation Title: “Data-Driven Framework for Forecasting Sedimentation at Culverts”
Focus Area: Hydroinformatics and Environmental Informatics
Passed the final thesis defense (on December 5th, 2018)

- Received multiple travel grants and research fellowship from NSF, UCGIS, and CGRER
- UI SETP (the University of Iowa Student Employee of the Year) Nominee

Graduate Certificate in Geoinformatics, 2014 to 2016
The University of Iowa, Iowa City, IA
Cumulative GPA: 3.94/4.00

MS, Civil and Environmental Engineering 2013 - 2015
The University of Iowa, Iowa City, IA
Cumulative GPA: 4.00/4.00
Dissertation Title: "Prototype Hydroinformatics-based System for Supporting Decision Making in Culvert Design and Monitoring"

- Received a Full-tuition Scholarship

B.S.E., Civil Engineering 2009 to 2013
The University of Iowa, Iowa City, IA
Cumulative G.P.A.: 3.54/4.00; Dean's List
Thesis Title: "Search Engine for Iowa Watershed Data Information System"

- *Graduated with Honors* in Civil Engineering

PROFESSIONAL EXPERIENCE

UNSW Sydney March 2025 - Present
Lecturer/Research Fellow (Level B)
School of Built Environment

Oak Ridge National Laboratory June 2021 – March 2025
Research Scientist, Computational Urban Sciences Group
Computational Sciences and Engineering Division

University of Tennessee, Knoxville Septmber 2024 - Septmber 2025
Adjunct Associate Professor
Department of Industrial and Systems Engineering (ISE)

Oak Ridge National Laboratory June 2019 – May 2021
Postdoctoral Research Associate, Computational Urban Sciences Group
Computational Sciences and Engineering Division

University of Iowa August 2013 – May 2019
Graduate Research Assistant
IIHR – Hydroscience & Engineering, Iowa City, Iowa

Institute of Rural Research and Development (IRRAD) December 2013 – January 2014
Teaching Assistant & Engineering Surveyor & GIS Specialist
Gurgaon, India

HONORS AND AWARDS

2024 Best Research Paper Award from the 51st International Conference on Computers & Industrial Engineering (CIE51), UNSW, Australia

2022 Best Poster Award from the 19th IEEE International Conference on Mobile Ad-hoc and Smart Systems (MASS)

2022 Award for the Most Creative Demonstration from IBPSA HackSimBuild 2022 competition in conjunction with Building Performance Analysis Conference and SimBuild 2022

2021 Invited presenter at the 1st GeoScale Workshop

2020 Significant Event Award (SEA) – ORNL, USDOE

Release of Real-time Ctwin v1.1 -- The Chattanooga Digital Twin for Mobility/Transportation

2018 Wayne L. Paulson Scholarship Award (1000 USD)

2018 Graduate College Post-Comprehensive Research Fellowship (9000 USD)

2017 UCGIS & NSF Travel Awards (2000 USD)

2017 CGRER Travel Award (600 USD)

2017 UI SETP (the University of Iowa Student Employee of the Year) Nominee

2016 Invited presenter at the United States Environmental Protection Agency (USEPA) headquarters

2016 Best Student Oral Presentation Award from the American Water Resources Association (AWRA)

2015 CGRER Travel Award (750 USD)

2013 Study Abroad International Student Merit-based Scholarship

2013 Segal Family Foundation Scholarship

2013 The University of Iowa Engineering College Merit-based Scholarship

PROFESSIONAL MEMBERSHIP

- Institute of Electrical and Electronics Engineers (IEEE)
- Association for Computing Machinery (ACM)
- Association of American Geographers (AAG)
- Chi-Epsilon National Civil Engineering Honor Society
- International Association for Hydro-Environment Engineering and Research (IAHR) Web Master of SIIHR and Iowa Young Professionals Network

JOURNAL PUBLICATION

1. **Xu, H.,** Zlatanova, S., Liang, R., & Canbulat, I. (2025). Generative AI as a Pillar for Predicting 2D and 3D Wildfire Spread: Beyond Physics-Based Models and Traditional Deep Learning. Fire, *8*(8), 293.
2. Yuan, J., Li, W., Wang, C., LaClair, T., **Xu, H.,** Berres, A., Kadav, P., Wang, H., Chen, B., Lim, H., Groelke, B., Shao, Y., Ekti, A. R., Chen, J., Wang, Q., & Sanyal, J. (in revision). *Real-world integration of traffic signal and connected automated vehicle speed control for energy-efficient mobility. Nature Communications (in revision)*.
3. **Xu, H.,** Sun, Y., Tupayachi, J., Omitaomu, O., Zlatanova, S., & Li, X. (2025). Towards the Autonomous Optimization of Urban Logistics: Training Generative AI with Scientific Tools via Agentic Digital Twins and Model Context Protocol. *arXiv preprint arXiv:2506.13068*.
4. **Xu, H.,** & Yu, X. Y. (2025). From powerpoint ui sketches to web-based applications: Pattern-driven code generation for gis dashboard development using knowledge-augmented llms, context-aware visual prompting, and the react framework. *arXiv preprint arXiv:2502.08756*.
5. **Xu, H.,** Boyaci, A., Lian, J., & Wilson, A. (2025). Explainable ai for multivariate time series pattern exploration: Latent space visual analytics with temporal fusion transformer and variational autoencoders in power grid event diagnosis. IEEE Access.
6. **Xu, H.,** Yuan, J., Zhou, A., Xu, G., Li, W., & Ye, X. (2024). GenAI-powered Multi-Agent Paradigm for Smart Urban Mobility: Opportunities and Challenges for Integrating Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) with Intelligent

Transportation Systems. Urban Human Mobility, 123-137.

7. **Xu, H.,** Omitaomu, F., Sabri, S., Zlatanova, S., Li, X., & Song, Y. (2024). Leveraging generative AI for urban digital twins: a scoping review on the autonomous generation of urban data, scenarios, designs, and 3D city models for smart city advancement. Urban Informatics, 3(1), 29.

8. Tupayachi, J., **Xu, H*.,** Omitaomu, O. A., Camur, M. C., Sharmin, A., & Li, X. (2024). Towards Next-Generation Urban Decision Support Systems through AI-Powered Construction of Scientific Ontology Using Large Language Models—A Case in Optimizing Intermodal Freight Transportation. Smart Cities, 7(5), 2392-2421.

9. Moriano, P., Berres, A., **Xu, H.,** & Sanyal, J. (2024). Spatiotemporal Features of Traffic Help Reduce Automatic Accident Detection Time. Expert Systems With Applications, 244, 122813.

10. Shi, L., Qu, M., Liu, L., Venegas, T. P., Wang, L., Dong, J., Cui, B., **Xu, H.,** Liu, X., & Li Y. (2024). Performance evaluation of underground thermal storage integrated dual-source heat pump systems. Energy and Buildings: 114349.

11. Berres, A., Moriano, P., **Xu, H.,** Tennille, S., Smith, L., Storey, J., & Sanyal, J. (2024). A traffic accident dataset for Chattanooga, Tennessee. Data in brief, 55, 110675.

12. **Xu, H.,** Shao, Y., Chen, J., Wang, C., & Berres, A. (2023). Semi-Automatic Geographic Information System Framework for Creating Photo-Realistic Digital Twin Cities to Support Autonomous Driving Research. Transportation Research Record, 03611981231205884.

13. **Xu, H.,** Yuan, J., Berres, A., Shao, Y., Wang, C., Li, W., Laclair, T., Sanyal, J., & Wang, H. (2023). A Mobile Edge Computing Framework for Traffic Optimization At Urban Intersections Through Cyber-Physical Integration. IEEE Transactions on Intelligent Vehicles, doi: 10.1109/TIV.2023.3332256.

14. **Xu, H.,** Berres, A., Yoginath, S. B., Sorensen, H., Nugent, P. J., Severino, J., ... & Sanyal, J. (2023). Smart Mobility in the Cloud: Enabling Real-Time Situational Awareness and Cyber-Physical Control Through a Digital Twin for Traffic. IEEE Transactions on Intelligent Transportation Systems.

15. **Xu, H.,** Berres, A. S., Shao, Y., Wang, C., J, New., Omitaomu, O. A. (2022). Towards a Smart Metaverse City: Immersive Realism and 3D Visualization of Digital Twin Cities. In Surya Durbha (Eds.). Advances in Scalable and Geospatial Analytics: New Trends, Challenges and Applications.

16. Muste, M., & **Xu, H*.** (2022). Multi-pronged approach for monitoring sedimentation processes at multi-barrel culverts. Journal of Hydrology, 610, 127840.

17. **Xu, H.,** Berres, A., Liu, Y., Allen-Dumas, M. R., & Sanyal, J. (2022). An overview of visualization and visual analytics applications in water resources management. Environmental Modelling & Software, 105396.

18. **Xu, H.,** Berres, A., Thakur, G., Sanyal, J., & Chinthavali, S. (2021). EPIsembleVis: A geo-visual analysis and comparison of the prediction ensembles of multiple COVID-19 models. Journal of Biomedical Informatics, 124, 103941.

19. **Xu, H.,** Berres, A., Tennille, S. A., Ravulaparthy, S. K., Wang, C., & Sanyal, J. (2021). Continuous Emulation and Multiscale Visualization of Traffic Flow Using Stationary Roadside Sensor Data. IEEE Transactions on Intelligent Transportation Systems.

20. **Xu, H.,** Wang, C., Berres, A., LaClair, T., & Sanyal, J. (2022). Interactive web application for traffic simulation data management and visualization. Transportation research record, 2676(1), 274-292.

21. Li, X., **Xu, H.,** Huang, X., Guo, C. A., Kang, Y., & Ye, X. (2021). Emerging geo-data sources to reveal human mobility dynamics during COVID-19 pandemic: opportunities and challenges. Computational Urban Science, 1(1), 1-9.
22. Berres, A. S., **Xu, H.,** Tennille, S. A., Severino, J., Ravulaparthy, S., & Sanyal, J. (2021). Explorative visualization for traffic safety using adaptive study areas. Transportation research record, 2675(6), 51-69.
23. Allen-Dumas, M. R., **Xu, H*.,** Kurte, K. R., & Rastogi, D. (2021). Toward urban water security: broadening the use of machine learning methods for mitigating urban water hazards. Frontiers in Water, 2, 562304.
24. Berres, A. S., LaClair, T. J., Wang, C., **Xu, H.**, Ravulaparthy, S., Todd, A., ... & Sanyal, J. (2020). Multiscale and Multivariate Transportation System Visualization for Shopping District Traffic and Regional Traffic. Transportation Research Record, 0361198120970526.
25. **Xu, H.,** Demir, I., Koylu, C., & Muste, M. (2019). A web-based geovisual analytics platform for identifying potential contributors to culvert sedimentation. Science of the Total Environment, 692, 806-817.
26. **Xu, H.,** Windsor, M., Muste, M., & Demir, I. (2020). A web-based decision support system for collaborative mitigation of multiple water-related hazards using serious gaming. Journal of environmental management, 255, 109887.
27. **Xu, H.,** Muste, M., & Demir, I. (2019). Web-based geospatial platform for the analysis and forecasting of sedimentation at culverts. Journal of Hydroinformatics, 21(6), 1064-1081.
28. Basnet, K., Muste, M., Constantinescu, G., Ho, H., & **Xu, H.** (2016). Close range photogrammetry for dynamically tracking drifted snow deposition. Cold Regions Science and Technology, 121, 141-153.

GRANT SUPPORTS

1. United States Department of Energy - Advanced Research Projects Agency-Energy (ARPA-E): $1,900,000
   Title: A Cognitive Freight Transportation Digital Twin for Resiliency and Emission Control Through Optimizing Intermodal Logistics (RECOIL)
   Role: Co-Principal Investigator (Co-PI)
   Duration: Fiscal Year 2023-Present
   Project URL: https://recoil.utk.edu/
   URL: https://arpa-e.energy.gov/technologies/exploratory-topics/intermodal-freight
2. United States Department of Energy - Nuclear Safety Research and Development (NSRD): $500,000
   Title: ML-Assisted Atmospheric Hazard Modeling for Effective Geospatial Risk Analysis
   Role: Co-Investigator
   Duration: Fiscal Year 2023-Present
3. United States Department of Energy - Vehicle Technologies Office (VTO): $4,000,000
   Title: Scaling up the Realtime Data, Simulation and Artificial Intelligence (AI) and Control for Optimizing Regional Mobility
   Role: Task Lead for developing Digital Twins
   Duration: Fiscal Year 2021-2023
   URL: https://www.energy.gov/eere/vehicles/articles/regional-mobility-chattanooga
4. The Iowa Department of Transportation - The Iowa Highway Research Board (IHRB)
   Title: Forecasting and Mitigation of Sedimentation at Culverts
   Role: Co-Investigator
   Duration: 01/2015-12/2015

URL:
https://publications.iowa.gov/27032/1/TR%20665_Final%20Report_Mitigation%20of%20Sedimentation%20at%20Multi-Box%20Culverts.pdf

REFEREED CONFERENCE PROCEEDINGS

1. Li, X., **Xu, H.,** Tupayachi, J., Omitaomu, O., & Wang, X. (2024). Empowering Cognitive Digital Twins with Generative Foundation Models: Developing a Low-Carbon Integrated Freight Transportation System. *The* 51*st International Conference on Computers & Industrial Engineering (*CIE51*),* December 11th, Sydney, NSW, Australia. arXiv preprint arXiv:2410.18089.
2. **Xu, H.,** Li, X., Tupayachi, J., Lian, J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and retrieval-augmented generation (RAG): A pilot study in semantic and contextual search for customized literature characterization for high-impact urban research. *In The 2nd ACM SIGSPATIAL International Workshop on Advances in Urban AI*, November 1, 2024, Atlanta, GA, USA. ACM.
3. **Xu, H.,** Liu, X., Dong, J., & Lian, J. (2024). A Digital Twin Approach to Support Real-time Situational Awareness and Intelligent Cyber-physical Control in Energy Smart Buildings. In Digital Innovation Center Of Excellence (DICE) - Idaho National Laboratory (INL), Idaho Falls, Idaho, May 1, 2024.
4. Niloofar, P., Lazarova-Molnar, S., Omitaomu. O., **Xu, H.,** & Li, X. (2023). A General Framework for Human-in-the-loop Cognitive Digital Twins. In 2023 Winter Simulation Conference. San Antonio, Texas, December 10-13, 2023.
5. Shao, Y., Wang, C., Berres, A., Yoshioka, J., Cook, A., & **Xu, H.** (2022). Computer vision-enabled smart traffic monitoring for sustainable transportation management. In International Conference on Transportation and Development 2022 (pp. 34-45).
6. **Xu, H.,** Berres, A., Shao, Y., and Wang, C. (2022). Using Photo-Realistic Immersive Virtual Reality Digital Twin and Visualization to Support Road Safety Training, Transportation Research Board (TRB) - Visualization Symposium, National Academy of Sciences.
7. Berres, A., **Xu, H.,** Wang, C., Wang Q., Ugirumurera, J., Jones, W., and Sanyal, J. (2022). Multivariate Spatiotemporal Visual Representations for Analyzing Traffic Performance at Signalized Intersections, Transportation Research Board (TRB) - Visualization Symposium, National Academy of Sciences.
8. **Xu, H.,** Liu, X., Lian, J., Li, Y., Malhotra, M., & Omitaomu, O. A. (2022, November). A geo-visual analysis for exploring the socioeconomic benefits of the heating electrification using geothermal energy. In Proceedings of the 5th ACM SIGSPATIAL International Workshop on Advances in Resilient and Intelligent Cities (pp. 11-15).
9. **Xu, H.,** Yuan, J., Wang, C., Shao, Y., Berres, A., and Laclair, T. (2022). A Mobile App for Intersectional Traffic Optimization through Real-Time Vehicle-to-Infrastructure (V2I) Communication and Cyber-Physical Control. IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2022). Denver, CO.
10. Chinthavali, S., Hasan, S.M., Yoginath, S., Xu, H., Nugent, P., Jones, T., Engebretsen, C., Olatt, J., Tansakul, V., Christopher Carter., Polsky An, Y., (2022). Alternative Timing and Synchronization Approach for Situational Awareness and Predictive Analytics, IEEE IRI 2022, Las Vegas, NV, USA. Resolution.
11. **Xu, H.,** Wang, C., Berres, A., Shao, Y., Yoshioka, J. (2022). Digital Twin and Real-time Situational Awareness for Sustainable and Smart Campus Management, AAG Annual Meeting 2022, New York, NY.

12. **Xu, H.,** Berres, A., Wang, C., Laclair, T. (2021). An Urban Informatics-based Vehicle-to-Infrastructure Communication Approach for the Smart Mobility and Environmental Management, AGU Fall Meeting, New Orleans, LA.
13. **Xu, H.,** Berres, A., Wang., C., LaClair, T., and Sanyal, J. (2021). Visualizing Vehicle Acceleration and Braking Energy at Intersections along a Major Traffic Corridor, ACM 2nd Workshop on Energy Data Visualization (EnergyVis II).
14. **Xu, H.,** Wang, C., Berres, A, LaClair, T. and Sanyal, J. (2021). RyThMiCCS: An Interactive Web-based Content Management System for Sharing and Comparing Traffic Simulation Outputs, Transportation Research Board Annual Meeting 2021.
15. **Xu, H.,** Sanyal., J., Berres, A. and Liu, Y. (2020). An Overview of Visual Analytics Applications in Water Resources Management, AWRA Annual Meeting, 2020.
16. Thakur, G., Sparks, K., Berres, A., Tansakul, V., Chinthavali, S., Whitehead, M., Schmidt, E., **Xu, H.,** Fan, J., Spears, D. and Cranfill, E., (2020), November. COVID-19 Joint Pandemic Modeling and Analysis Platform. In Proceedings of the 1st ACM SIGSPATIAL International Workshop on Modeling and Understanding the Spread of COVID-19 (pp. 43-52).
17. **Xu, H.,** Sanyal., J., Berres, A. and Tennille, S. (2019). Optimization of Network Datasets for Web-based System using Composite Bezier Curves, AAG Annual Meeting 2020, Denver, Co.
18. Ugirumurera, J., Severino, J.A., Wang, Q., Ravulaparthy, S., Berres, A., Nugent, P., Sorensen, H., Moore, A., Todd, A., Nag, A. and Tennille, S., Peterson, S., Potter., K., **Xu, H.,** Jones. W., Sanyal., J. (2020). High Performance Computing Traffic Simulations for Real-Time Traffic Control of Mobility in Chattanooga Region (No. NREL/PO-2C00-75009). National Renewable Energy Lab.(NREL), Golden, CO.
19. **Xu, H.,** Berres, A. and Sanyal., J. (2019). A Client-side Web Application for Visualizing Massive Regional Mobility Data Collected from Real-time Traffic Sensors, AGU Fall Meeting 2019, San Francisco, CA
20. Muste, M., **Xu, H.** and Mocanu, M. (2017). Community Engagement in Water Resources Planning Using Serious Gaming, IEEE 13th International Conference on Intelligent Computer Communication and Processing," September 7-9, 2017, Cluj-Napoca, Romania.
21. **Xu, H.,** Hameed, H., Windsor, M., Muste, M., Demir, I., Smith, J., Hunemuller, T., Stevenson, M. B. (2017). Decision-support System for Underpinning Collaborative Planning for Multi-hazard Mitigation," Proceeding of the 37th IAHR World Congress, August 13-18, 2017, Kuala Lumpur, Malaysia.
22. **Xu, H.,** Hameed, H., Windsor, Smith, J., Hunemuller, T., M., Muste, M., Demir, I., Stevenson, M. B. (2017). An Iowa Prototype for a Generic Watershed Decision Support System, 11th Annual Iowa Water Conference, March 22-23, 2017, Ames, IA.
23. Muste, M. **Xu, H.,** Goetz, V., Claman, D., Abu-Hawash, A. (2017). Sedimentation Mitigation at Multi-box Culverts, 96th Annual Meeting, Transportation Research Board, January 8-12, 2017, Washington D.C.
24. Garousi-Nejad. I., Li. Xiao., Petrasova. A., and **Xu, H.** (2016). High-resolution Flood Inundation Mapping Using National Water Model Forecasts for Emergency Management, UCGIS symposium.
25. **Xu, H.,** Hameed, H., Demir, I., Muste, M. (2016). Visualization Platform for Collaborative Modelling, 2016 AWRA Summer Specialty Conference - GIS and Water Resources IX, American Water Resources Association, July 11-13, 2016, Sacramento, CA. –Awarded with the Best Student Oral Presentation for Haowen Xu.

26. **Xu, H.,** Hameed, H., Shen, B., Demir, I., Muste, M., Stevenson, M.B. and Hunemuller, T. (2016). Prototype Decision Support System for Interjurisdictional Collaboration in Water Resource Management, August 21-26, 2016, Hydroinformatics Conference, Incheon, Korea.
27. Muste, M., Cheng, Z., Firoozfar, A.R., Tsai, H-W., Loeser, T. and Xu, H. (2016). Impacts of Unsteady Flows on Monitoring Stream Flows, River Flow Conference, July 12-15, 2016, St. Louis, MO, USA.
28. **Xu, H.,** Shen, B. and Muste, M. (2015). Geo-portal for Sustainable Culvert Design and Monitoring, Proceedings 36th IAHR World Congress, 28 June – 3 July, The Hague, The Netherlands.
29. **Xu, H.,** Amir, A.A., Muste, M. (2014). Web-based Engine for Discovery of Observations Using Landscape Units," 11th International Conference on Hydroinformatics, New York City, NY.

TECHNICAL REPORTS AND WHITE PAPERS

1. **Xu, H.,** Berres, A. S., Yoginath, S., Kurte, K, Paleti, R., New, J., Sanyal. J. (2022). Towards Adaptive Decision Support: A Perspective from Intelligent and Annotated Visual Analytics for Exploring Big Urban Mobility Data, ASCR Workshop on Visualization for Scientific Discovery, Decision-Making, & Communication. US Department of Energy.
2. **Xu, H.,** Berres, A. S., Yoginath, S., Sanyal. J. (2022). Metaverse for Urban Informatics: A Solution towards Participatory Smart City Applications, ASCR Workshop on the Science of Scientific-Software Development and Use. US Department of Energy.
3. Muste, M. and **Xu, H.** (2020). Development of self-cleaning box culverts, IIHR Report No. TR-719, Submitted to the Iowa Highway Research Board, Ames, Iowa.
4. Muste, M. and **Xu, H.** (2017). Mitigation of Sedimentation at Multi-Box Culverts, IIHR Report No. TR-655, Submitted to the Iowa Highway Research Board, Ames, Iowa.
5. Muste, M. and **Xu, H.** (2017). Sedimentation Mitigation using Streamlined Culvert Geometry, Report ST-001, Iowa Department of Transportation, Statewide Transportation Innovation Council, Federal Highway Administration, McLean, VA.
6. Riedinger, M., Ceynar, D., Eichinger, W., Muste, M., Plenner, S., Tsai, H.W., Willis, W. and **Xu, H.,** (2015). Salt Spray Measurement by Lidar, IIHR Report No. 482, Submitted to the Federal Highway Administration, Washington, D.C.

PROFESSIONAL CERTIFICATES

- IBM Professional Certificate - Machine Learning with Python (R96QPR685KBM)
- IBM Professional Certificate - Introduction to Deep Learning & Neural Networks with Keras (T43PD6PEWMJE)

PROFESSIONAL SERVICES

Guest Editor of Special Journal Issue

- Journal Special Issue "Frontiers in Urban Water Infrastructure", WATER, 2021 https://www.mdpi.com/journal/water/special_issues/urban_water_infrastructure

Student Advisory & Mentorship

- Sidney Ozcan (Undergraduate) Co-advised through 2023 U.S. Department of Energy - Science Undergraduate Laboratory Internships (SULI) program
  Research Outcome: *Earthquake Vulnerability Assessment of California 's Petroleum DIstribution Network (ORNL SULI Poster)*
- Jill Sturtevant (Ph.D Student at Baylor University) through 2023 NSF Internships Program
  Research Topic: *Understanding Traffic in and around Yellowstone National Park*

after the June 2022 Flooding
- Jose Tupayachi (Ph.D Student at the University of Tennessee) through the ARPA-E RECOIL Project

Academic Services

- Grant Awardee and Exhibator, U.S. Department of Energy, ARPA-E Energy Innovation Summit 2024, Dallas, Texas, May 22-24, 2024
- Program Committee Member, ISPRS Technical Commission IV Symposium 2024
- Workshop Co-Organizer, IWCTS 2023 - The 16th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2023), Hamburg Germany, November 13, 2023
- Program Committee Member, Visualization in Environmental Sciences Workshop (EnvirVis) - EUROVIS 2023 - 25th EG Conference on Visualization, Leipzig, Germany, June 12-13, 2023
- Workshop Co-Organizer, IWCTS 2022 - The 15th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2022), Seattle, WA, November 1, 2022
- Program Committee Member, ARIC 2022 - The 5th ACM SIGSPATIAL International Workshop on Advances in Resilient and Intelligent Cities, Seattle, WA, USA, November 1, 2022
- Session Chair - Session "Geographic Information Analysis and Computing Technologies in Ecosystem Science" at AAG Annual Meeting 2022
- Session Co-organizer - Session "Symposium on Human Dynamics Research: Emerging Urban-Geo Big Data in Transportation Research" at AAG Annual Meeting 2022
- Program Committee Member, IWCTS 2021 - The 14th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2021), Beijing, China, November 1, 2021
- Session Chair - Session "Geovisualization and Geovisual Analytics for Exploring Complex Urban Systems" at AAG Annual Meeting 2021
- Session Co-organizer - Session "Symposium on Human Dynamics Research: Emerging Urban-Geo Big Data in Transportation Research" at AAG Annual Meeting 2021

Referee Services

2024

- Accident Analysis & Prevention (1 Review)
- Geo-spatial Information Science (1 Review)
- EuroVis 2024 (2 Reviews)

2023

- Transportation Research Board Annual Meeting 2023 (3 Reviews)
- 1st ACM SIGSPATIAL International Workshop on Advances in Urban-A (1 Review)
- Accident Analysis & Prevention (1 Review)

- IEEE Transactions on Intelligent Transportation Systems (2 Reviews)
- EuroVis 2023 (4 Reviews)
- Open Research Europe - European Union (1 Review)
- European Journal of Soil Science (EJSS) (1 Review)
- Journal of Hydroinformatics (1 Review)

2022

- European Journal of Soil Science (EJSS) (1 Review)
- The 15th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2022) (3 Reviews)
- The 5th ACM SIGSPATIAL International Workshop on Advances in Resilient and Intelligent Cities (ARIC 2022) (2 Reviews)
- Transportation Research Board Annual Meeting 2022 (6 Reviews)
- MDPI Sustainability (1 Review)
- Social Sciences (1 Review)
- Journal of Hydroinformatics (1 Review)
- IEEE Transactions on Intelligent Transportation Systems (3 Reviews)
- IEEE Computer Graphics and Applications (1 Review)
- Frontiers In Water (1 Reviews)
- CATENA (1 Review)

2022

- The 14th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2022) (3 Reviews)
- Transportation Research Board Annual Meeting 2021 (3 Reviews)
- Applied Sciences (1 Review)
- IEEE VIS Short Paper (3 Reviews)
- Computational Urban Science (2 Reviews)
- Journal of Hydroinformatics (1 Review)
- Sustainability (3 Reviews)
- Water (1 Review)

2020

- Transportation Research Board Annual Meeting 2020 (2 Reviews)
- IEEE VIS Short Paper (1 Review)
- International Conference on Materials, Computer Engineering and Education Technology (2 Reviews)

- Energies (2 Reviews)
- Sustainability (4 Reviews)

2019

- Journal of Hydroinformatics (1 Review)
- Journal of Soils and Sediments (1 Review)